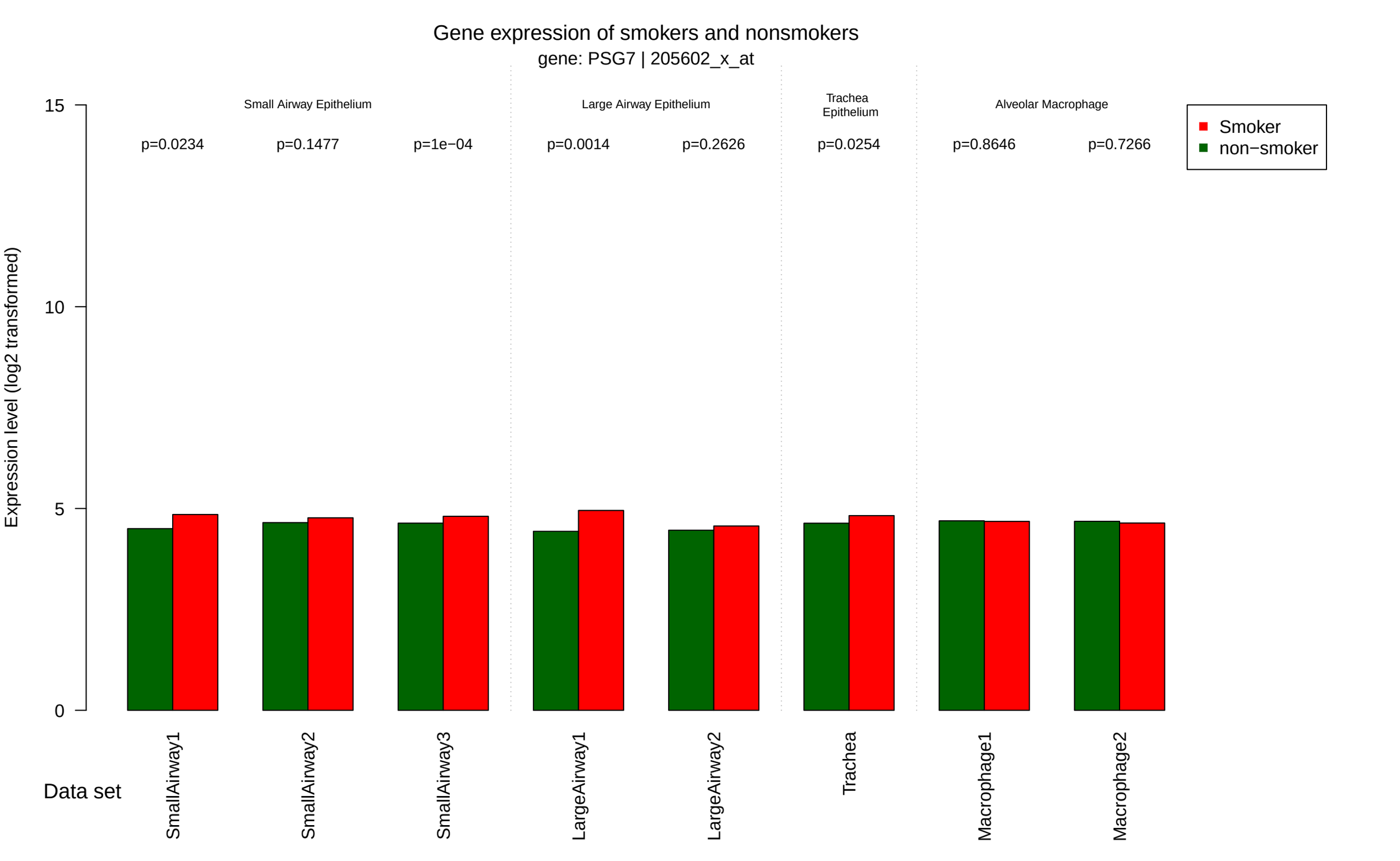
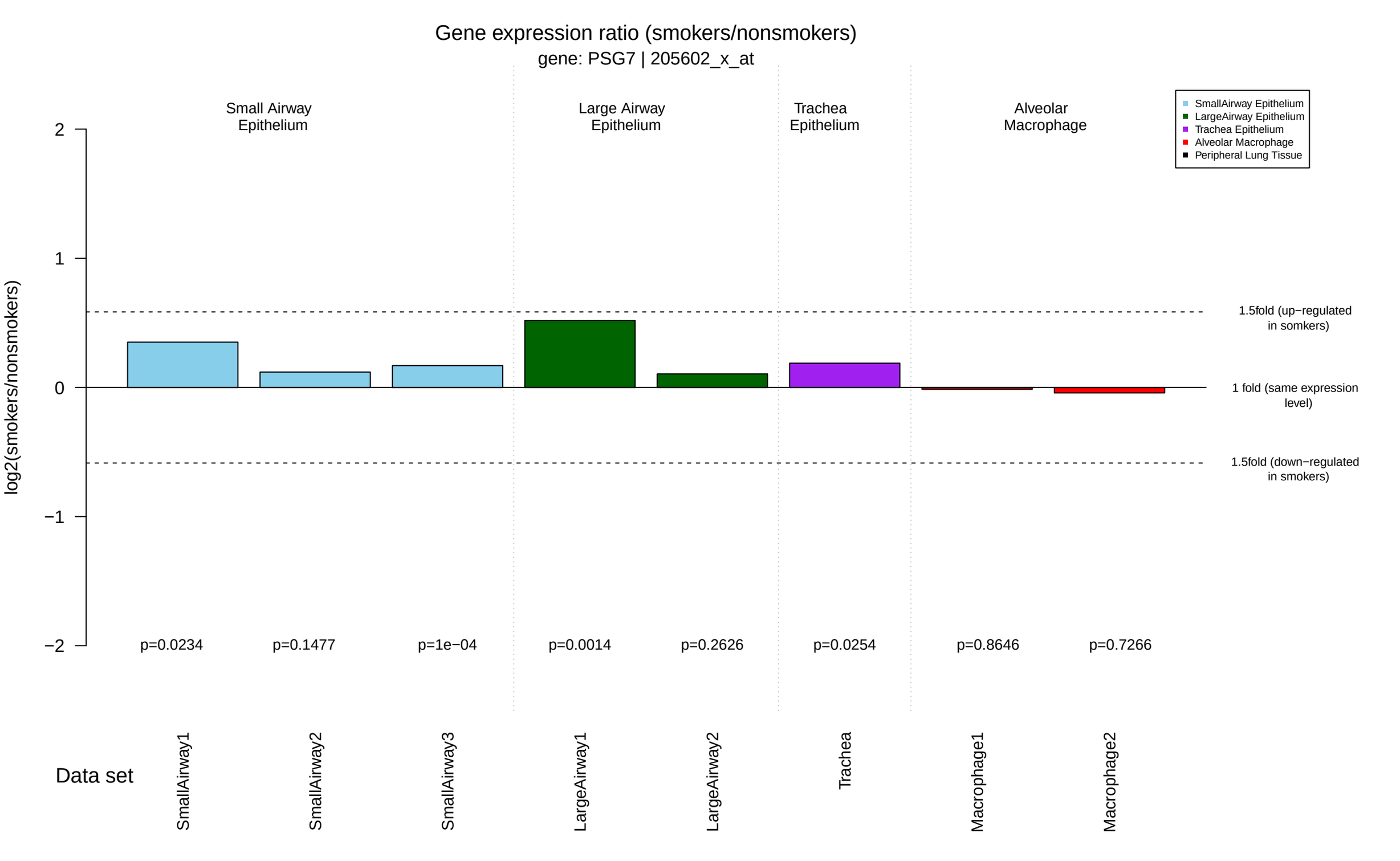
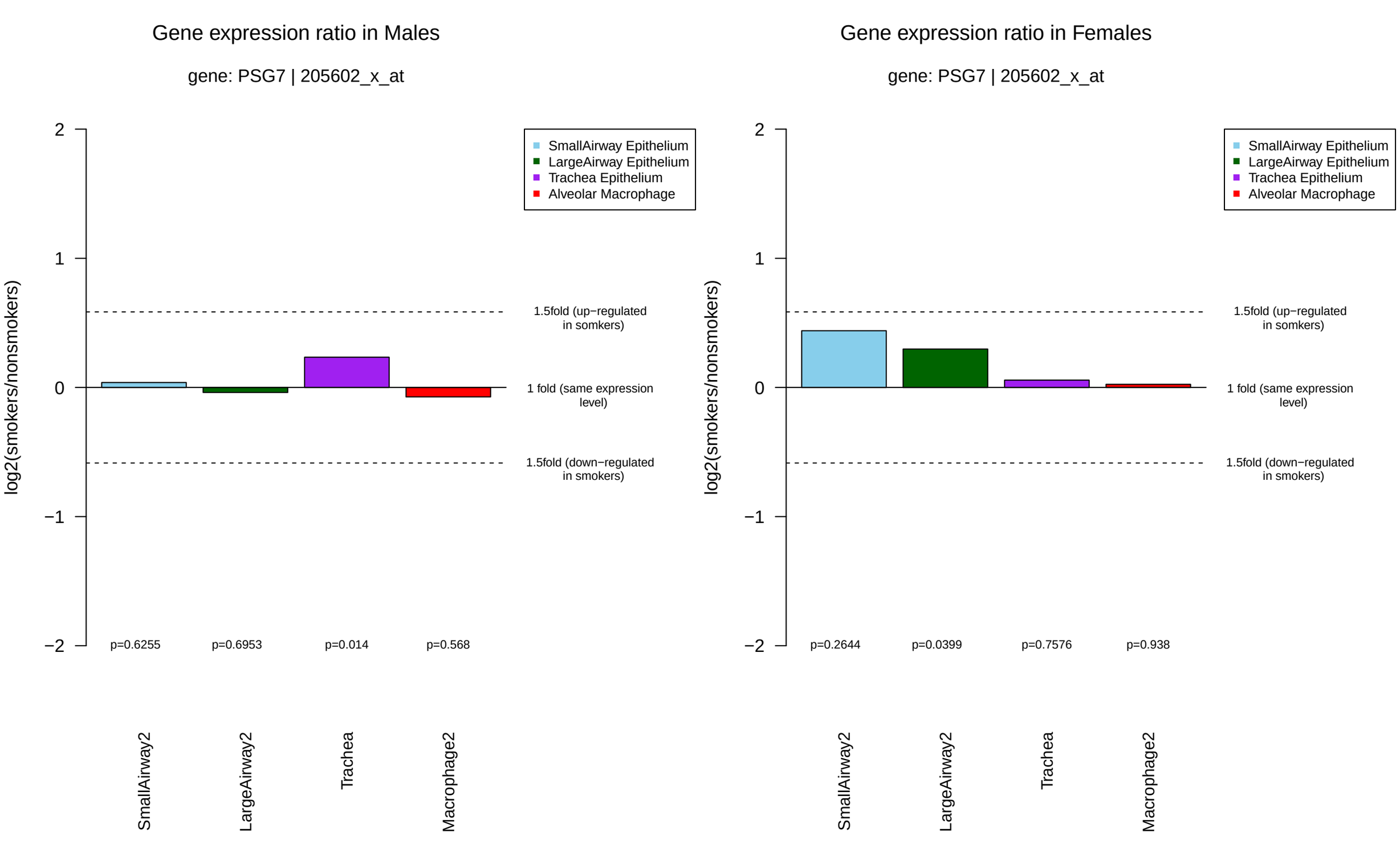
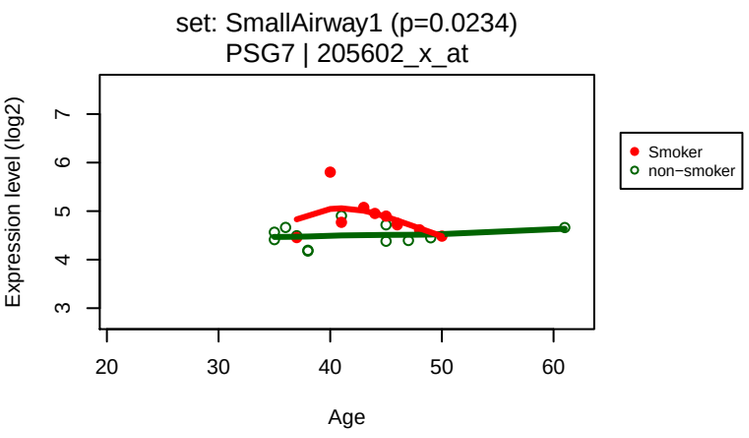
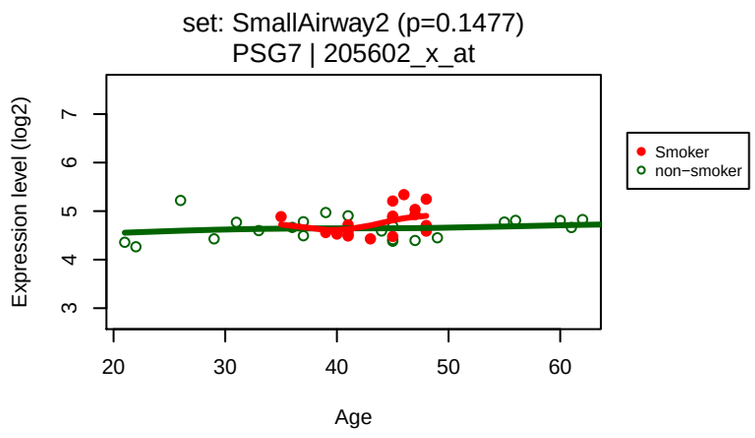
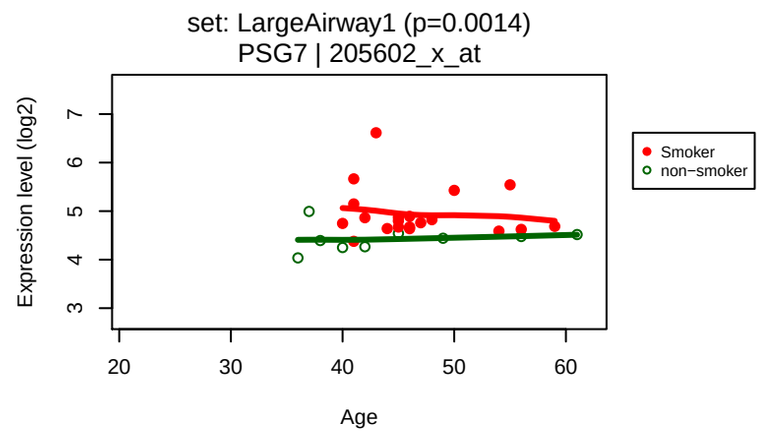

Gene expression of smokers and nonsmokers
gene: PSG7 | 205602_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0234
p=0.1477
p=1e−04
p=0.0014
p=0.2626
p=0.0254
p=0.8646
p=0.7266
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

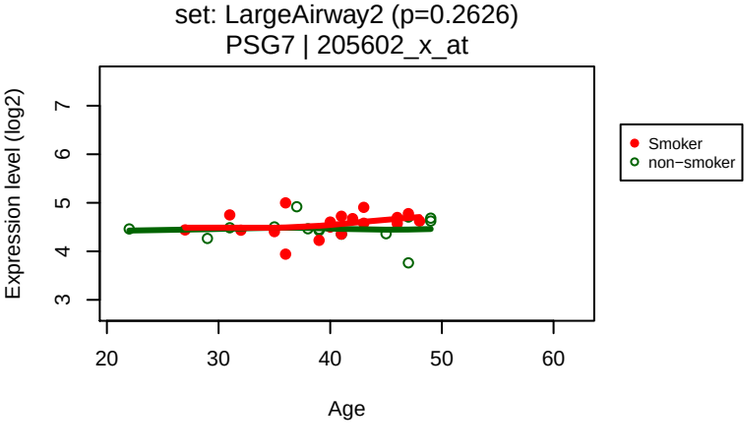

Gene expression ratio (smokers/nonsmokers)
gene: PSG7 | 205602_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0234
p=0.1477
p=1e−04
p=0.0014
p=0.2626
p=0.0254
p=0.8646
p=0.7266
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

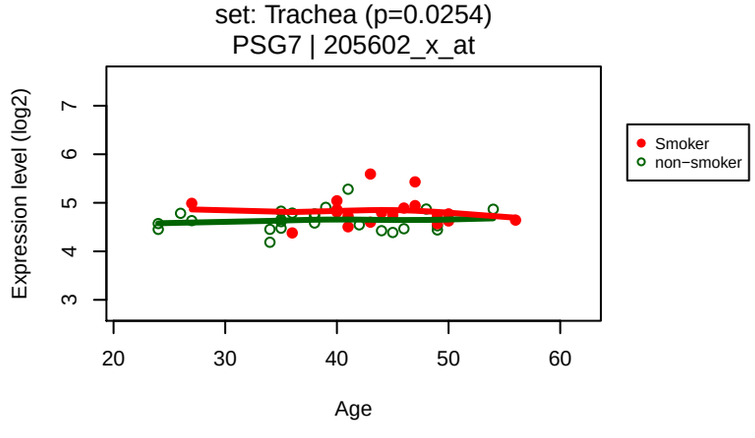

Gene expression ratio in Males
gene: PSG7 | 205602_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6255
p=0.6953
p=0.014
p=0.568
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PSG7 | 205602_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2644
p=0.0399
p=0.7576
p=0.938
SmallAirway2
LargeAirway2
Trachea
Macrophage2

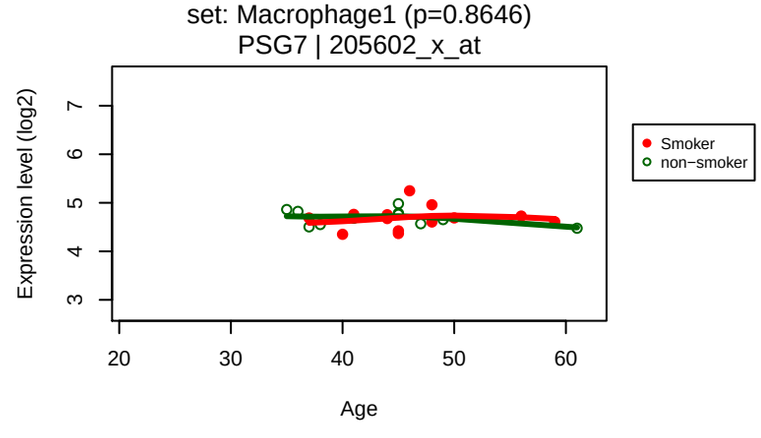

set: SmallAirway1 (p=0.0234)
PSG7 | 205602_x_at
Expression level (log2)
Age
Smoker
non−smoker

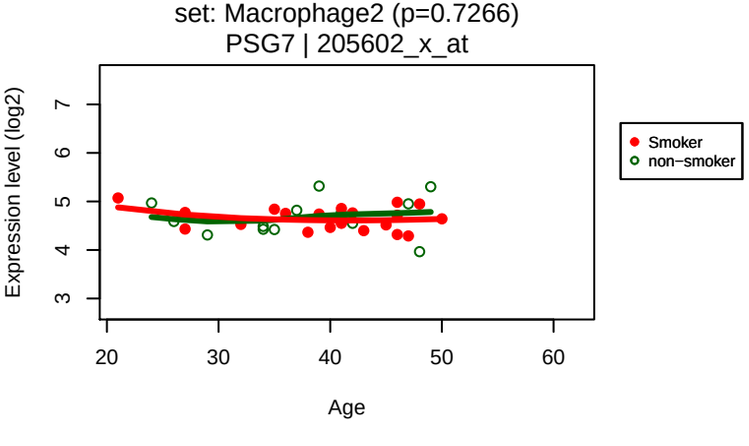

set: SmallAirway2 (p=0.1477)
PSG7 | 205602_x_at
Expression level (log2)
Age
Smoker
non−smoker

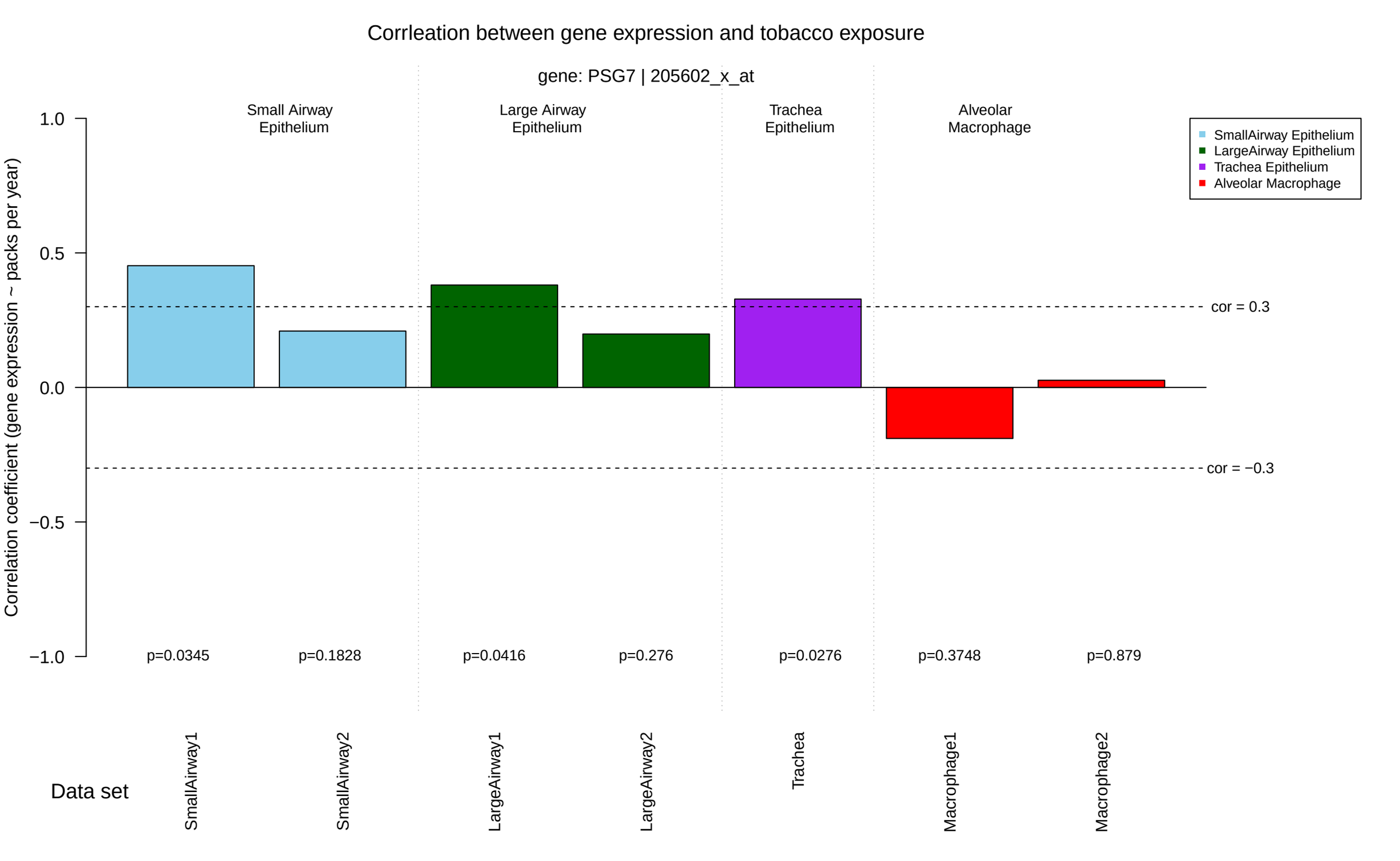

set: LargeAirway1 (p=0.0014)
PSG7 | 205602_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2626)
PSG7 | 205602_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0254)
PSG7 | 205602_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8646)
PSG7 | 205602_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7266)
PSG7 | 205602_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PSG7 | 205602_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0345
p=0.1828
p=0.0416
p=0.276
p=0.0276
p=0.3748
p=0.879
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2